

Autonomous Racing via Reinforcement Learning and Vision-Based Defense

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Abstract—This paper studies defensive blocking for two-agent autonomous racing in the F1TENTH setting. We propose a using reinforcement learning at the behavioral decision layer, while maintaining the classical methods for nominal path following, local RRT* replanning, and pure pursuit tracking. The ego vehicle learns blocking behavior from opponent-relative motion history and track context, and outputs continuous blocking parameters: lateral offset magnitude, hold time, and return-to-raceline rate. The policy is trained in simulation using Proximal Policy Optimization with randomized spawn conditions and a bimodal curriculum for robustness. Compared with a simple rule-based baseline, the learned policy achieves the same overall success rate while exhibiting more adaptive defensive behavior, including stronger recovery of the track-progress gap in several high-pressure scenarios.

Index Terms—Reinforcement learning, PPO, autonomous racing, defensive maneuvers, multi-agent systems, vision-based detection

I. INTRODUCTION

Autonomous racing requires more than simply following the fastest line. It compresses the core challenges of self-driving real-time perception, high-speed control, and adversarial multi-agent interaction into a competitive setting. Our project studies defensive driving in the F1TENTH setting, where two racecars interact on a shared track and one vehicle attempts to maintain its lead while another attempts to advance. In this setting, the ego vehicle must decide when to defend, how aggressive it needs to be, and when to return to the optimal race line.

While single-agent time-trial racing has seen strong results with RL agents matching optimal controllers [1], wheel-to-wheel racing remains underdeveloped. Our initial project proposal considered a larger behavior stack where an agent would consider between overtaking, blocking, and optimal path following. However, as the project evolved, we narrowed the scope to focus on learning defensive blocking in a two-agent environment. In the final version, the ego vehicle learns only blocking-related decisions, while path following and object avoidance remained classical. This lets us isolate whether reinforcement learning is genuinely useful for interactive defense compared to a rule-based defense algorithm.

II. PROBLEM STATEMENT

The goal of this project is to design a blocking policy for an ego F1TENTH vehicle racing against a single opponent in

simulation and then transfer the policy to the physical vehicle. The ego should maintain its lead when the opponent approaches from behind, but should avoid unnecessary or overly aggressive blocking that causes instability, wall collisions, or wasted distance. We essentially want a policy that maps the opponent's motion to our ego's decisions on blocking.

The problem is challenging because blocking is not a purely geometric task. A useful blocking policy will consider whether the opponent is truly trying to pass or if they are just probing. The policy should also understand how aggressive they should block and for how long. In addition, a good blocking policy would consider the track before making any decisions. A policy that blocks too early wastes time and may destabilize the vehicle, while a policy that blocks too late allows the opponent to establish overlap and complete a pass.

III. MOTIVATION

In Formula 1, 60% of position changes occur via overtaking but strategy and timing matter as much as raw speed [2]. Real F1 teams use ML and live telemetry for these decisions; we mirror that at the autonomy level. In F1tenths, the same principle holds true: A car that cannot defend will be passed and eventually will lose.

This project is also motivated by a control-design gap. It is relatively straightforward to hand-engineer a nominal controller, a local RRT-based bypass, or a fixed blocking offset. The harder question is when those behaviors should be used and how much defensive action is appropriate in context. We therefore focus on learning blocking behavior from opponent-relative state and track geometry while keeping the lower-level planner and controller fixed.

IV. RELATED WORK

Over the past few years there have been multiple reinforcement learning (RL) papers that show strong results in autonomous racing especially in places where aggressive control and multi-agent interaction make hand-designed objectives difficult to tune. Wurman et al. showed deep RL could outperform championship-level Gran Turismo drivers, validating RL as a credible framework for high-performance racing. However, GT Sophy relied on some privileged global external state information that is not directly available to a

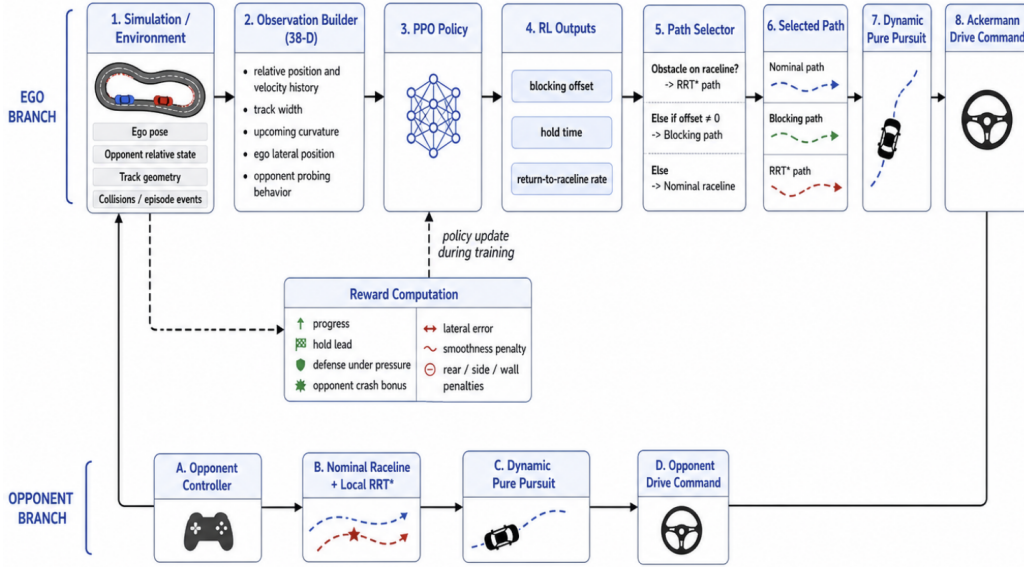


Fig. 1. Overall system RL pipeline

physical racecar. This limits its relevance to onboard defensive behavior on F1TENTH-scale hardware [3]. Vasco et al. later targets this dependency on global privileged sensing by adding a vision-based racing agent that used ego-centric camera input and onboard quantities. Their system focused on time-trial racing rather than wheel-to-wheel defensive interaction [4]. More recently, Lee et al. extended the vision-based RL to competitive Gran Turismo racing, indicating that onboard RL models are competitive and credible. However, this study does not look at using rear facing camera for blocking [5].

The second line of work compares learning-based control against modern model-based control in aggressive racing domains. Song et al. directly compared RL vs. MPC, showing RL’s fundamental advantages in dynamic scenarios. While this result strongly motivates the use of RL for dynamic competitive tasks, it does not address multi-agent car racing, rearward threat perception, or defensive maneuver selection [1]. Additionally, surveys of F1TENTH have indicated that the platform contains both a classical and learning-based methods, but the field is still dominated by waypoint following, trajectory optimization, and obstacle avoidance rather than learned defensive algorithms [6].

The closest prior works to ours are those that explicitly consider interaction with opponents. Ögretmen et al. applied RL to overtaking trajectories in multi-agent racing and showed that learned methods can outperform classical planners in multi-agent scenarios. Importantly, their paper also analyzes blocking-like opponent interaction, but focus more on how to overtake an opponent implementing defensive strategies [7]. For F1tenth, Steiner et al. reported recent work achieved 87% overtake success with opponent-trained RL vs. 56% for solo-trained. However, their emphasis is on overtaking rather than defensive blocking, and the method does not address rear-view threat detection [8].

Altogether, these works suggest three conclusions. First, RL is a strong candidate for racing tasks that involve behavior planning. Second, vision-based sensing is becoming realistic for onboard racing agents. Third, interactive racing papers still focus primarily on overtaking or trajectory generation. Our project targets the gap between these areas. We study defensive blocking in a two-agent F1TENTH setting, where a learned policy uses opponent-relative motion history and track context to decide how aggressive to block and how long to block for.

V. PROPOSED APPROACH

Our approach is to create a learned model that predicts opponent overtaking intent and blocks accordingly. The opponent vehicle uses a non-learning controller that follows the nominal raceline and uses local RRT* replanning only when needed to overtake. The ego vehicle uses pure pursuit tracking and switches between blocking and the optimal line base on the RL model. Reinforcement learning is used only at the behavior-selection layer for blocking as it determines the strength and timing of the defensive maneuver, while a fixed planner converts those parameters into a blocking trajectory. Figure 1 depicts the general pipeline we used for the project.

A. Why RL?

We use reinforcement learning because blocking is an interaction problem with hidden intent. A hard-coded defense rule can effectively detect distance and relative speed, but becomes increasingly complex to tune a single threshold-based controller that behaves well across all situations like straightaways, corners, different closing speeds, repeated side probing, and varying track widths. RL is attractive here because the correct action is not simply a deterministic function of one measurement. Instead, the agent must infer whether the opponent is going to overtake, which side the opponent will overtake from, and whether the ego should defend at

the current track location. We intentionally limit the use of RL in the pipeline. The optimal path, local RRT* logic, and Pure Pursuit controller remain classical. This makes the project easier to compare against baseline blocking methods and allows for a more stable model. The main question is whether the RL can outperform a rule based baseline for blocking.

B. Observation Space

The policy observes a compact state vector that combines opponent history and track context. The observation is constructed as

$$o_t = [h_t, e_t] \quad (1)$$

where h_t is a short history of opponent-relative motion and e_t is a vector of instantaneous geometric and track features. The history portion stores the last H samples of

$$[r_x, r_y, \dot{r}_x, \dot{r}_y] \quad (2)$$

where r_x and r_y are the opponent relative position to the ego and $\dot{r}_x, \dot{r}_y$ are the corresponding relative velocities. In our implementation, $H = 6$, so this contributes 24 features. The extra feature vector e_t adds

$$[r_x, r_y, d, v_{\text{close}}, \bar{\kappa}_{1m}, \kappa_{2m}^{\text{max}}, \rho_{\text{turn}}, w_{\text{track}}, e_{\text{lat}}, \Delta s, p_{\text{switch}}, p_{\text{side}}, v_{\text{ego}}, v_{\text{opp}}], \quad (3)$$

where d is relative distance, v_{close} is closing speed, $\bar{\kappa}_{1m}$ and κ_{2m}^{max} summarize upcoming curvature, ρ_{turn} is turn density, w_{track} is track width, e_{lat} is the lateral shift from the optimal line, Δs is the signed progress gap, and $p_{\text{switch}}, p_{\text{side}}$ indicate the opponent's probing behavior. This gives a total observation size of

$$4H + 14 = 38. \quad (4)$$

This observation design gives the model enough information to infer opponent intent while analyzing the track shape for a better judgment. In simulation, these quantities are built from ground-truth odometry, while in the real-car, the same structure can be estimated from a rear-facing vision module and vehicle localization.

C. Action Space

The policy creates a 3 dimensional continuous action space

$$a_t = [a_t^{\text{off}}, a_t^{\text{hold}}, a_t^{\text{return}}] \in [0, 1]^3 \quad (5)$$

These outputs are normalized and used in three blocking parameters:

$$\delta_{\text{block}} = a_t^{\text{off}} \delta_{\text{max}}, \quad (6)$$

$$T_{\text{hold}} = T_{\text{min}} + a_t^{\text{hold}}(T_{\text{max}} - T_{\text{min}}), \quad (7)$$

$$\lambda_{\text{return}} = \lambda_{\text{min}} + a_t^{\text{return}}(\lambda_{\text{max}} - \lambda_{\text{min}}). \quad (8)$$

Here, δ_{block} defines how aggressive the ego may block laterally off the optimal line. The T_{hold} controls how long the block is maintained for while being limited with a T_{min} and T_{max} time. Lastly, the λ_{return} controls how quickly the car returns to the raceline. To simplify the RL models problem, we

compute the sign of the blocking offset based on the opponent-side information. In this way, the RL model focuses just on determining the magnitude and length of the block rather than relearning how to block both the left and right sides.

This action design is stronger than a discrete behavior classifier because it lets the policy dynamically control the entire defensive maneuver continuously based on the track and opponent intent. It can produce small corrections on turns, larger offsets on straights, and shorter or longer holds depending on the threat.

D. Reward Function

The reward function is designed to encourage holding a favorable position while staying on the optimal line as long as possible. Let r_t denote the total reward at time t . The main components are:

- positive reward for forward progress ($r_{\text{prog}} = \Delta s_t$),
- positive reward for remaining ahead ($r_{\text{ego ahead}}$),
- a pressure-conditioned reward for using blocking when the opponent is genuinely threatening ($r_{\text{threat}} = p_t \bar{\delta}_t$),
- a bonus when the opponent collides on the track without making contact ($r_{\text{opp collision}}$),
- penalties for lateral deviation ($\bar{e}_{\text{lat},t}$), overly abrupt action changes ($r_{\text{abrupt action change}}$), being passed ($r_{\text{ego passed}}$), contact (r_{rear} and r_{side}), ego track collisions ($r_{\text{track collision}}$), and elapsed time (-0.002).

A simplified form is

$$r_t = 0.75 \Delta s_t + 0.20 r_{\text{ego ahead}} + 0.50 r_{\text{threat}} + 4.00 r_{\text{opp collision}} - 0.50 \bar{e}_{\text{lat},t} - 0.03 r_{\text{abrupt action change}} - 10.0 r_{\text{ego passed}} - 0.10 r_{\text{rear}} - 8.0 r_{\text{side}} - 15.0 r_{\text{track collision}} - 0.002. \quad (9)$$

This structure encodes the main tradeoff of the project. The agent should learn to block only the opponent is pressuring from behind and the agent should avoid any large unnecessary lateral motion that would pull it off the optimal line. This design prevents a bad solution where the ego blocks constantly regardless of context.

E. Training Setup

Training is performed in a two-agent F1TENTH simulation. The ego uses the RL-based blocking policy, while the opponent uses a non-learning controller that follows the nominal raceline with Pure Pursuit and relies on RRT* when needed. The simulator uses randomized initial conditions so that the ego experiences different track geometries and closing scenarios across episodes. Specifically, the spawn conditions can vary in longitudinal gap, lateral offset, and yaw, improving robustness to different attack situations.

We use PPO as the policy optimization algorithm. The observation space and action space are continuous, making PPO a good choice for simple and quick stable policy-gradient optimization. The simulation runs a two-agent racing environment in `f1tenth_gym` with fixed map, global waypoints, local controllers, and a blocking planner that transforms learned parameters into a path. This lets us train the decision layer

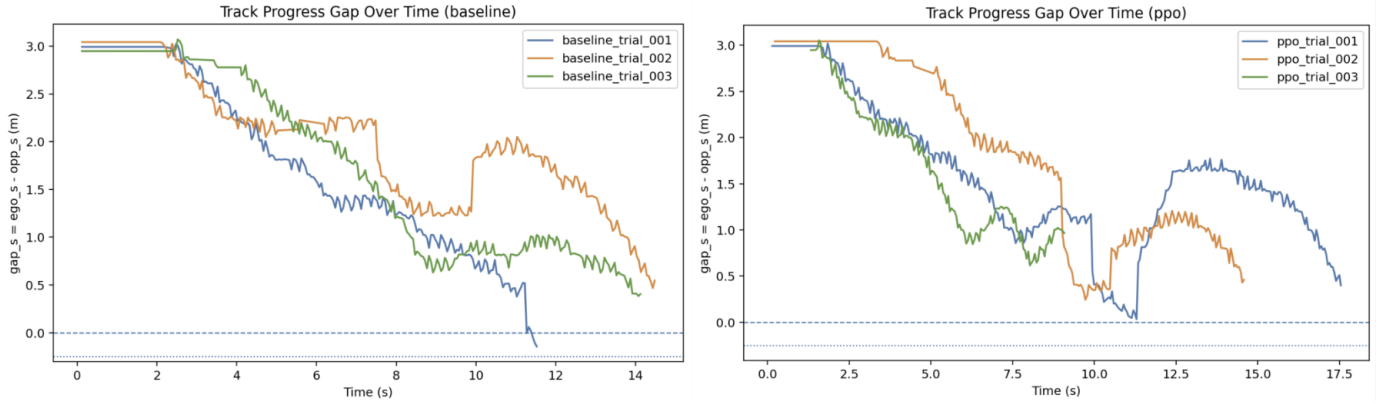


Fig. 2. Tracked progress gap over three trials for both methods

without changing the classical motion stack. Table I lists the parameters we used to train our PPO model in simulation. The bimodal spawn indicates a 70% chance the opponent will spawn in at a threatening gap (within 2 meters) and a 30% the opponent will spawn at a safer distance (from 3-6 meters). This is intended to help with the model learning to block with purpose.

TABLE I
TRAINING SETUP FOR THE PPO BLOCKING POLICY.

Setting	Value
<i>RL formulation</i>	
RL algorithm	PPO (Stable-Baselines3)
Observation space	38-dimensional state vector
Action space	3-D continuous: blocking offset, hold time, return-to-raceline rate
<i>Training hyperparameters</i>	
Total training timesteps	100,000
Spawn curriculum	Bimodal
Threatening spawns	70% threatening, 30% easy
Entropy coefficient	0.001
Batch size	128
Rollout steps (n_steps)	1024

F. YOLO Model Setup

For the simulation, we used ground-truth information to produce the observation space. For real life applicaiton, we designed a perception stack for real-car transfer. The real-car version uses a rear-facing RealSense camera and a custom YOLO model to detect the opponent from color images, combine that detection with depth measurements from the camera, estimate the observation space, and then construct the same observation layout used in simulation.

VI. RESULTS

Here is a table

Both the Pure Pursuit baseline and PPO achieve a 100% success rate, indicating that the task is reliably solved in sim-

TABLE II
COMPARISON OF BASELINE PURE PURSUIT AND PPO PERFORMANCE.

Method	Success (%)	Duration (s)	Threat (s)	Block (s)	Response (s)
Baseline (PP)	100	13.37	6.92	8.41	0.83
PPO	100	13.73	7.97	5.50	1.13

ulation regardless of the control strategy. Consequently, performance differences are better interpreted through efficiency and behavioral characteristics rather than task completion alone.

In terms of performance, the baseline achieves a slightly lower average duration (13.37 s compared to 13.73 s for PPO), suggesting a marginal advantage in maintaining overall pace. This is consistent with the deterministic and stable nature of Pure Pursuit, which follows a predefined trajectory with minimal deviation.

The PPO agent exhibits a higher average threat time (7.97 s vs. 6.92 s), indicating that it operates under opponent pressure for longer periods. This suggests a more reactive defensive strategy, where the agent allows closer proximity before responding. In contrast, the baseline maintains longer block times (8.41 s vs. 5.50 s), demonstrating more consistent but less adaptive defensive positioning.

Additionally, the baseline controller shows faster response times (0.83 s vs. 1.13 s), highlighting its ability to react quickly due to its simple control structure. The slower response observed in PPO may stem from policy inference latency or more selective defensive triggers.

Figure 2 compares the track progress gap over time for the baseline and PPO blocking agents across 3 trials. In both methods, the ego begins with an initial lead of approximately 3 m, and the gap generally decreases as the opponent applies pressure from behind. The baseline controller shows smoother behavior, indicating a stable but relatively rigid defense strategy. In contrast, the PPO agent exhibits greater variability, where there are multiple sharp drops and subsequent recoveries in the progress gap. Notably, PPO allows the opponent to approach more closely in several trials, but in two cases it

restores a substantially larger positive gap after the opponent nearly reaches the ego. This recovery is likely due to the ego holding its position around a turn while the opponent takes a less optimal race line to try and overtake. This suggests that the learned policy is more reactive and adaptive, though less consistent than the rule-based baseline.

Overall, while PPO matches the baseline in success rate, it trades off responsiveness and defensive persistence for a more flexible and adaptive strategy. Meanwhile, the rule-based baseline remains more stable and reactive, but lacks the adaptability to dynamically adjust its behavior under varying opponent interactions.

VII. CONCLUSION AND FUTURE WORK

This paper presented a reinforcement learning-based blocking policy for autonomous racing on the FITENTH platform. By restricting RL to the behavioral decision layer and retaining classical planning and control for trajectory tracking, we isolate the impact of learning in interactive defensive scenarios. The proposed PPO-based policy learns when and how aggressively to block based on the relative state of the opponent and the context of the track.

Experimental results show that the learned policy achieves a 100% success rate, matching the rule-based defensive baseline in maintaining race outcomes. However, the behavioral differences between the two approaches are significant. The baseline exhibits more stable, consistent, and faster responses due to its deterministic structure, while the PPO agent demonstrates more adaptive and context-dependent defensive behavior. In particular, the learned policy allows for closer opponent interaction, but is capable of recovering and maintaining position in challenging scenarios, highlighting its ability to react to dynamic race conditions.

These results suggest that reinforcement learning provides value not in basic task completion, but in shaping more flexible and context aware behaviors that are difficult to encode with fixed rules. At the same time, the tradeoff between adaptability and consistency remains a key challenge, as the learned policy introduces variability and slightly slower response times.

In the future, we will focus on improving both the robustness and scope of the proposed system for better sim-to-real transfer and applicability on the track. This can be addressed through improved reward function tuning and implementing noise in the simulation. We can further improve the model by including randomness in the speed of the ego vehicle and opponent vehicle to make it applicable for high speed racing. Future work may lean to building a unified decision-making framework that jointly handles offensive and defensive behaviors, allowing for more complex race strategies beyond blocking alone.

In addition, perception can be strengthened by refining the training of the YOLO based detection model using a larger and more diverse dataset. Extending the framework to multi-opponent scenarios will allow evaluation in more realistic racing environments where interactions are more complex than 1v1 settings. Finally, benchmarking against a more complex

rule-based defensive algorithm baseline in full head-to-head race scenarios will provide a more comprehensive comparison of performance, consistency, and strategic behavior.

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